

Real Analysis HW 5

Robert O'Connell

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BS - Page 86 Exercise 7

Let $\{x_n\}$ be a Cauchy sequence such that x_n is an integer for every $n \in \mathbb{N}$
Show that $\{x_n\}$ is ultimately a constant sequence

$\{x_n\}$ Cauchy $\Rightarrow \forall \epsilon > 0, |x_m - x_n| < \epsilon \forall m, n \geq N$
Pick $\epsilon = \frac{1}{2}$

$$\begin{aligned} |x_m - x_n| &< \frac{1}{2} \\ -\frac{1}{2} &< x_m - x_n < \frac{1}{2} \end{aligned}$$

We know that the difference of any two integers is an integer, say $x_m - x_n = k, k \in \mathbb{Z}$

$$\begin{aligned} -\frac{1}{2} &< k < \frac{1}{2} \\ \Rightarrow k &= 0 \end{aligned}$$

i.e. $\forall m, n \geq N, x_m - x_n = 0 \Rightarrow x_m = x_n$
 $\Rightarrow \{x_n\}$ is ultimately a constant sequence

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If $0 < r < 1$ and $|x_{n+1} - x_n| < r^n$ for all $n \in \mathbb{N}$
Show that $\{x_n\}$ is a Cauchy sequence

$$\begin{aligned} |x_n - x_{n-1}| &< r^{n-1} \\ -r^{n-1} + x_{n-1} &< x_n < r^{n-1} + x_{n-1} \end{aligned}$$

Then repeating this, estimating for x_{n-1} ,

$$-r^{n-1} - r^{n-2} + x_{n-2} < x_n < r^{n-1} + r^{n-2} + x_{n-2}$$

Then for $m < n$

$$\begin{aligned} -(r^{n-1} + r^{n-2} + \dots + r^m) + x_m &< x_n < r^{n-1} + r^{n-2} + \dots + r^m + x_m \\ |x_n - x_m| &< r^{n-1} + r^{n-2} + \dots + r^m \\ &= r^m \frac{1 - r^{n-1-m}}{1 - r} \\ &< r^m \frac{1}{1 - r} \end{aligned}$$

We know r_m converges to 0, hence $\frac{r^m}{1-r}$ also converges to 0, since $1 - r$ is a constant ($\neq 0$)
i.e. $\forall \epsilon > 0 \exists N \in \mathbb{N}$ such that $|\frac{r^m}{1-r}| < \epsilon \forall n \geq N$

$$\Rightarrow \forall \epsilon > 0, |x_n - x_m| < \epsilon \forall m \geq N, n > m$$

For $n = m$ it is obvious that $\forall \epsilon > 0, |x_m - x_m| = 0 < \epsilon$
Thus

$$\Rightarrow \forall \epsilon > 0, |x_n - x_m| < \epsilon \forall m, n \geq N$$

Hence $\{x_n\}$ Cauchy

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If $\sum a_n$ with $a_n > 0$ is convergent, then is $\sum a_n^2$ always convergent? Either prove it or give a counterexample

Since $\sum a_n$ convergent $\Rightarrow \lim_{n \rightarrow \infty} a_n = 0$ and $a_n > 0 \forall n$

$$\Rightarrow \exists N \text{ such that } a_n < 1 \forall n \geq N$$

If $a_n < 1 \Rightarrow a_n^2 < a_n < 1$

$$\Rightarrow \exists N \text{ such that } 0 < a_n^2 < a_n < 1 \forall n \geq N$$

We have that the convergence of series only depends on the tail of the series

i.e. $\sum_{n=1}^{\infty} x_n$ converges $\iff \sum_{n=N}^{\infty} x_n$ converges

and also that if $0 < x_n < y_n$ and if $\sum y_n$ converges then $\sum x_n$ converges

Hence $\sum a_n^2$ converges

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If $\sum a_n$ with $a_n > 0$ is convergent, then is $\sum \sqrt{a_n}$ always convergent? Either prove it or give a counterexample

Counterexample:

Consider $a_n = \frac{1}{n^2}$, $\sum \frac{1}{n^2}$ is a p-series with $p = 2$

Thus $\sum a_n$ converges

Now consider $\sqrt{a_n} = \frac{1}{n}$, we know $\sum \frac{1}{n}$ does not converge

Hence $\sum \sqrt{a_n}$ does not always converge if $\sum a_n$ converges